Lab 7 : Azure Networking with Jumpbox & Private VM

Steps;

- VNet
- Public + Private subnets
- Route tables (public + private)
- Internet route in the public RT
- Jumpbox (AlmaLinux) with public IP
- Private VM (AlmaLinux) with no public IP
- Access private VM via jumpbox

STEP 1: Create Resource Group

1. Go to Resource groups → Create
2. Name:
3. <studentname>-rg

Example: raj-rg

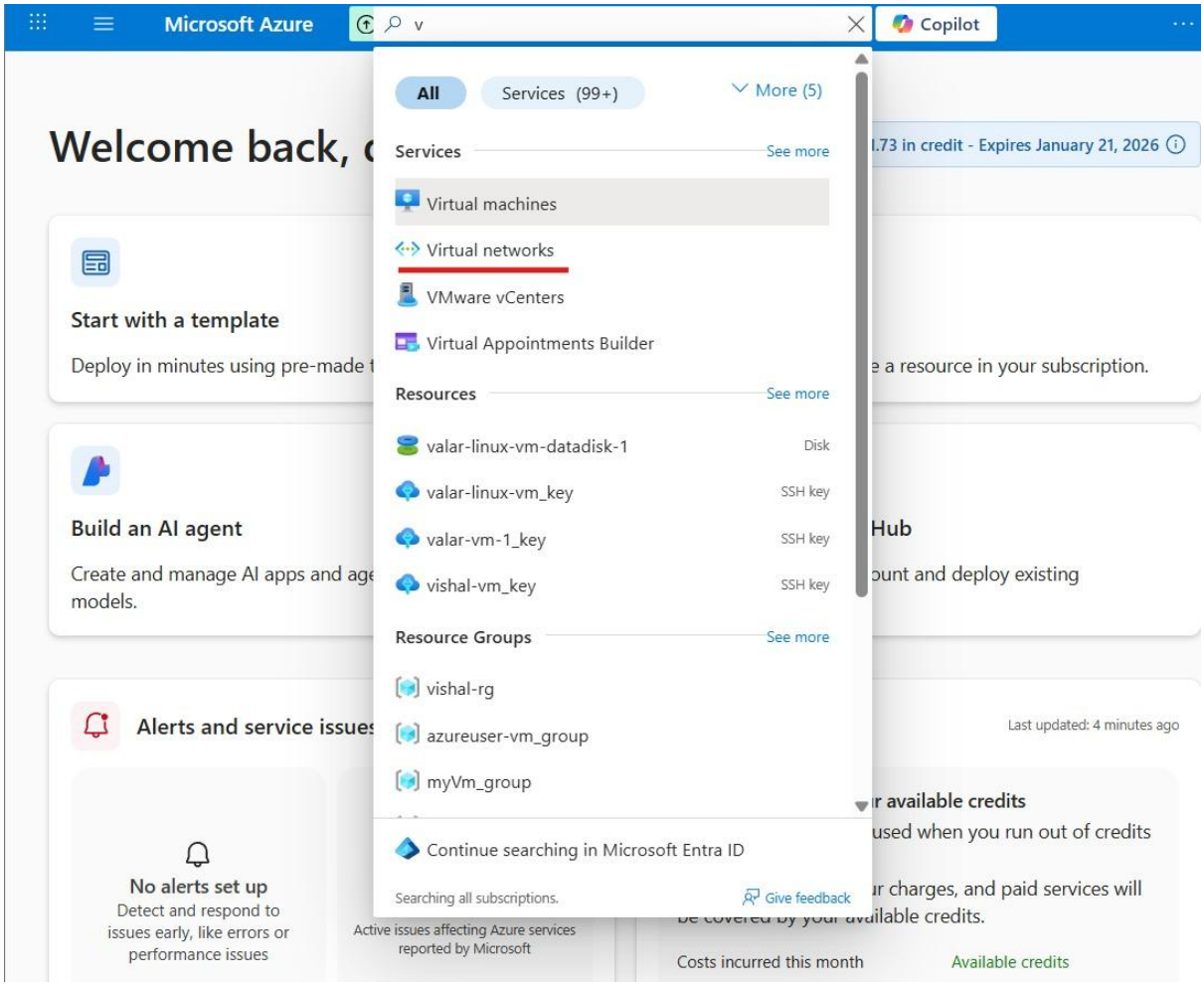
4. Select region: East US
5. Review + Create → Create

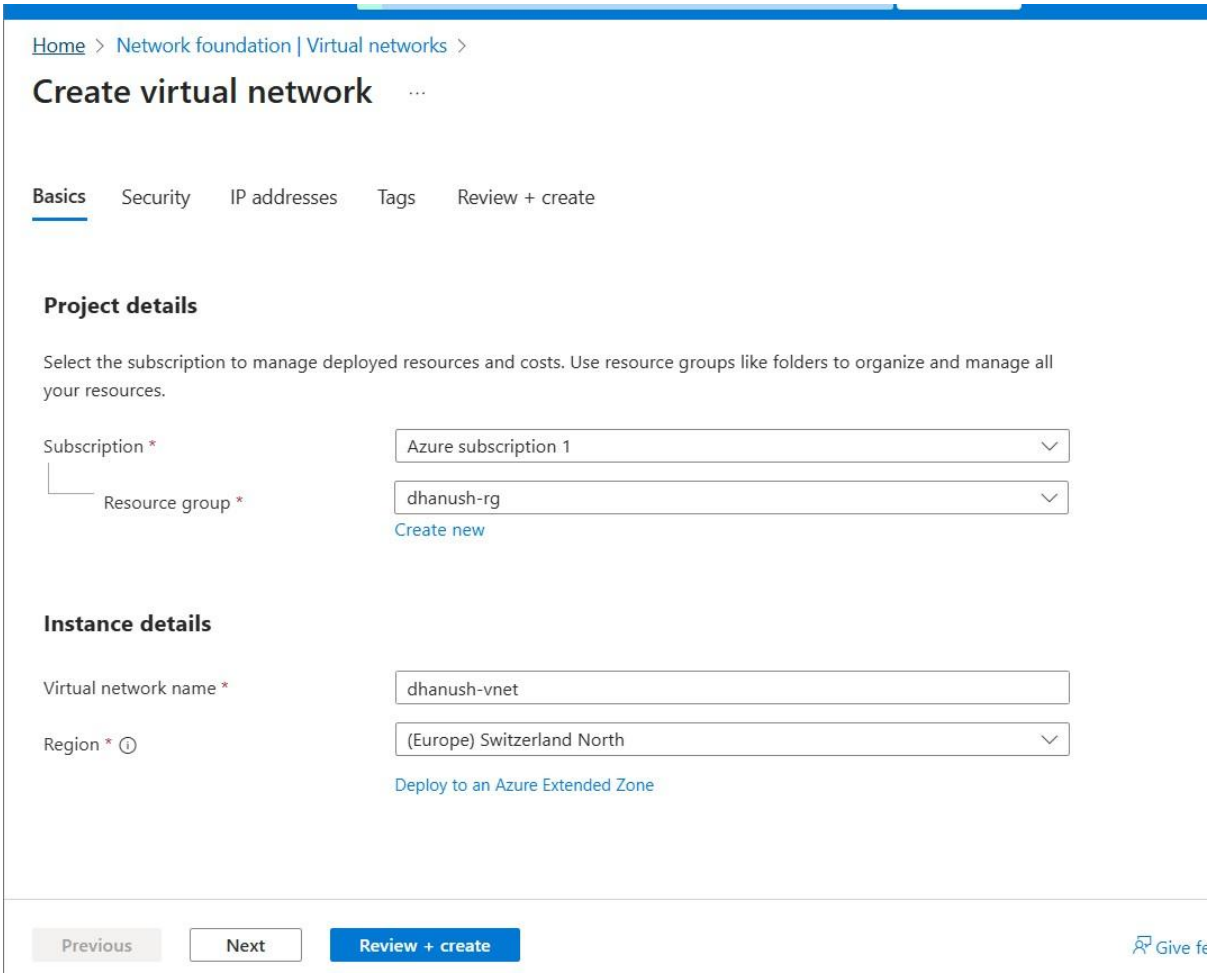
STEP 2: Create Virtual Network

1. Go to Virtual networks → Create
2. Name:
3. <studentname>-vnet

Example: raj-vnet

4. Address space: 10.0.0.0/16
5. Add subnets:
    - Public subnet
      → 10.0.1.0/24 Name: <studentname>-public-subnet
    - Private subnet
      → 10.0.2.0/24 Name: <studentname>-private-subnet
6. Review + Create → Create


Microsoft Azure
v
Copilot
All
Services (99+)
More (5)
Welcome back,
Services
See more
Virtual machines
Virtual networks
VMware vCenters
Virtual Appointments Builder
Resources
See more
valar-linux-vm-datadisk-1
Disk
valar-linux-vm_key
SSH key
valar-vm-1_key
SSH key
vishal-vm_key
SSH key
Resource Groups
See more
vishal-rg
azureuser-vm_group
myVm_group
Continue searching in Microsoft Entra ID
Searching all subscriptions.
Give feedback
.73 in credit - Expires January 21, 2026
Start with a template
Deploy in minutes using pre-made
e a resource in your subscription.
Build an AI agent
Create and manage AI apps and ag
models.
Hub
ount and deploy existing
Alerts and service issues
Last updated: 4 minutes ago
No alerts set up
Detect and respond to issues early, like errors or performance issues
Active issues affecting Azure services reported by Microsoft
r available credits
used when you run out of credits
r charges, and paid services will
ailable credits.
Costs incurred this month
Available credits



DELETE THE DEAFULT IP'S

Home > Network foundation | Virtual networks >

Create virtual network ...

Basics Security **IP addresses** Tags Review + create

assigns the resource an IP address from the subnet. Learn more

☐ Allocate using IP address pools. Learn more

+ Add a subnet

10.0.0.0/16 Delete address space

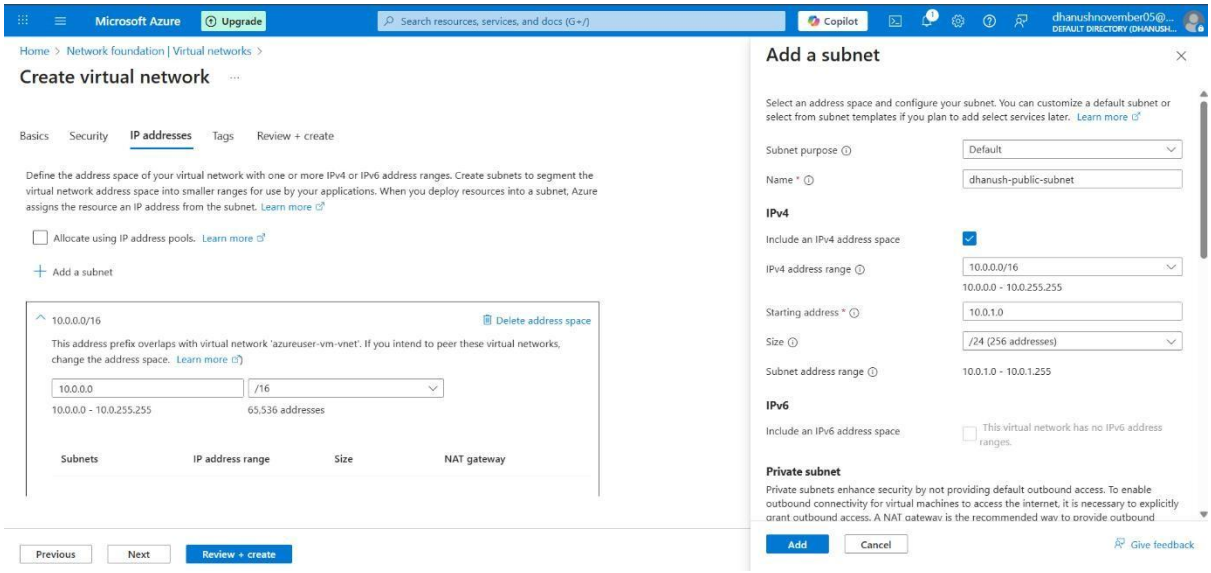
This address prefix overlaps with virtual network 'azureuser-vm-vnet'. If you intend to peer these virtual networks, change the address space. Learn more

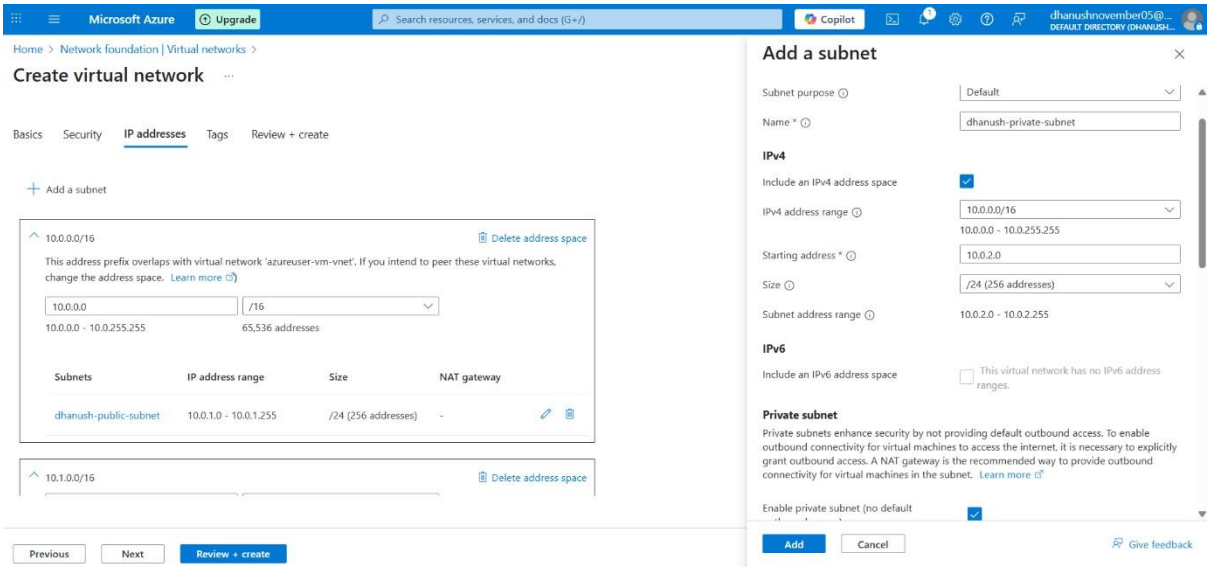
10.0.0.0 /16

10.0.0.0 - 10.0.255.255 65,536 addresses

Subnets	IP address range	Size	NAT gateway
default	10.0.0.0 - 10.0.0.255	/24 (256 addresses)	-

Previous Next **Review + create** Give





Home > Network foundation | Virtual networks >

Create virtual network ...

Basics Security **IP addresses** Tags Review + create

+ Add a subnet

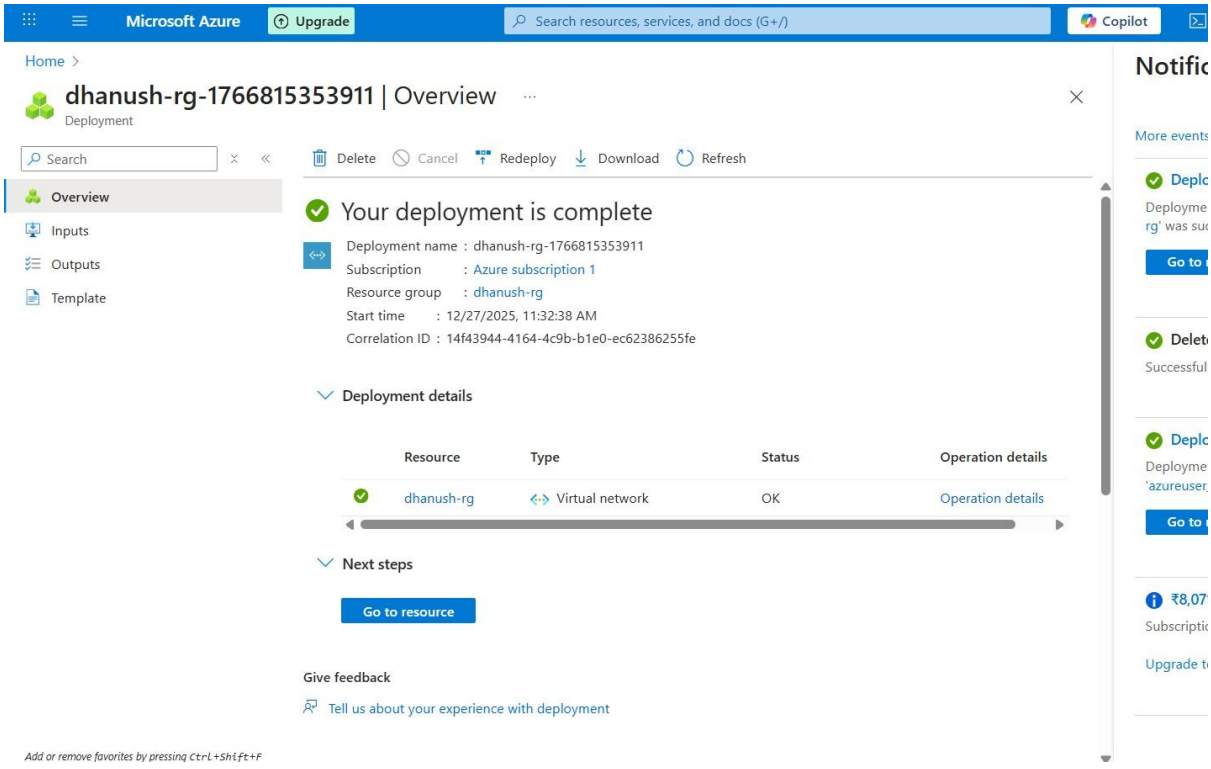
10.0.0.0/16 Delete address space

This address prefix overlaps with virtual network 'azureuser-vm-vnet'. If you intend to peer these virtual networks, change the address space. Learn more

10.0.0.0 | /16

10.0.0.0 - 10.0.255.255 65,536 addresses

Subnets	IP address range	Size	NAT gateway
dhanush-public-subnet	10.0.1.0 - 10.0.1.255	/24 (256 addresses)	-
dhanush-private-subnet	10.0.2.0 - 10.0.2.255	/24 (256 addresses)	-



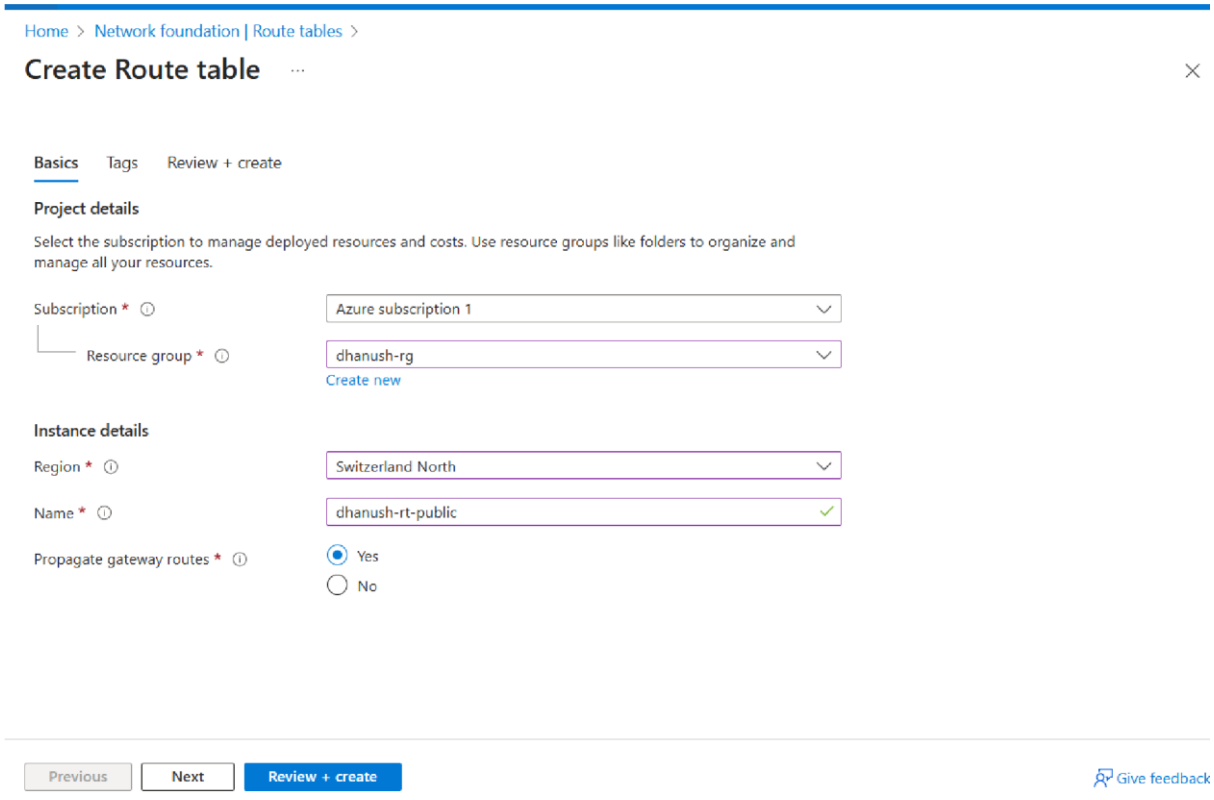
STEP 3: Create Route Tables

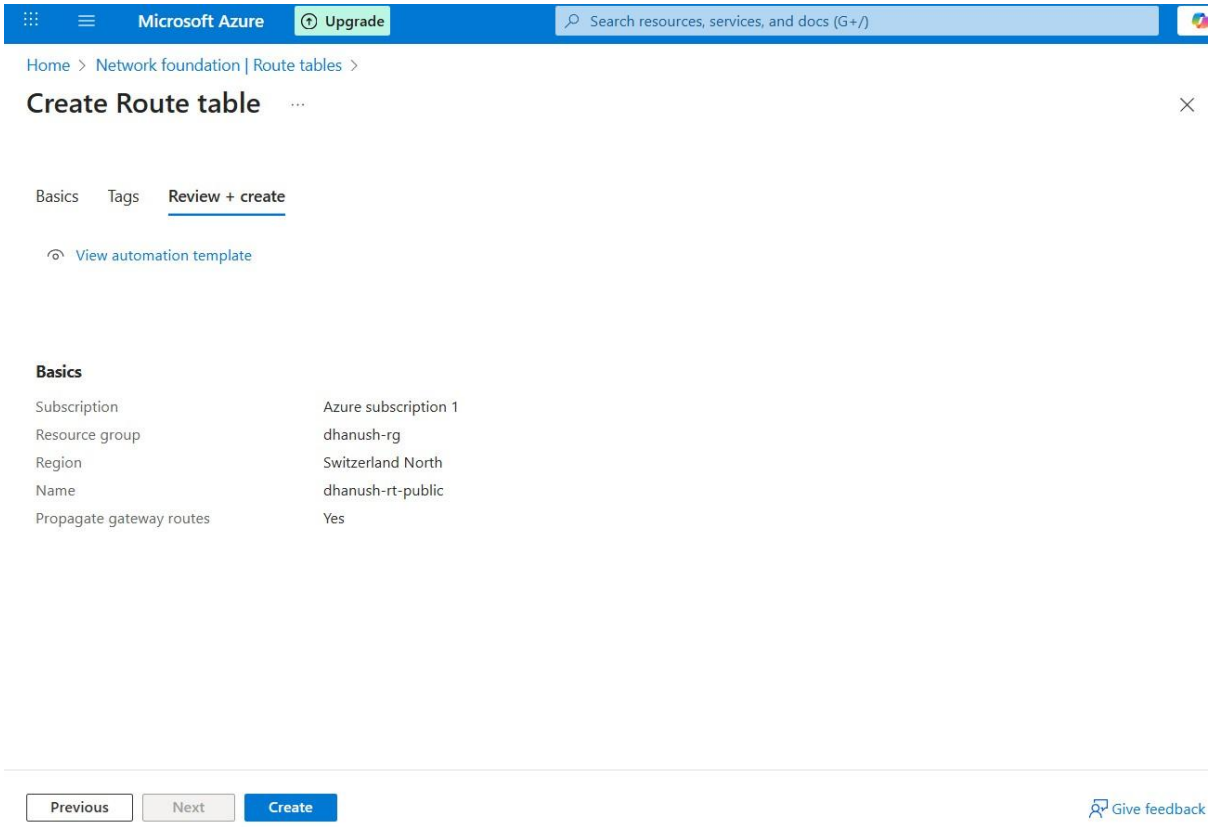
Public Route Table

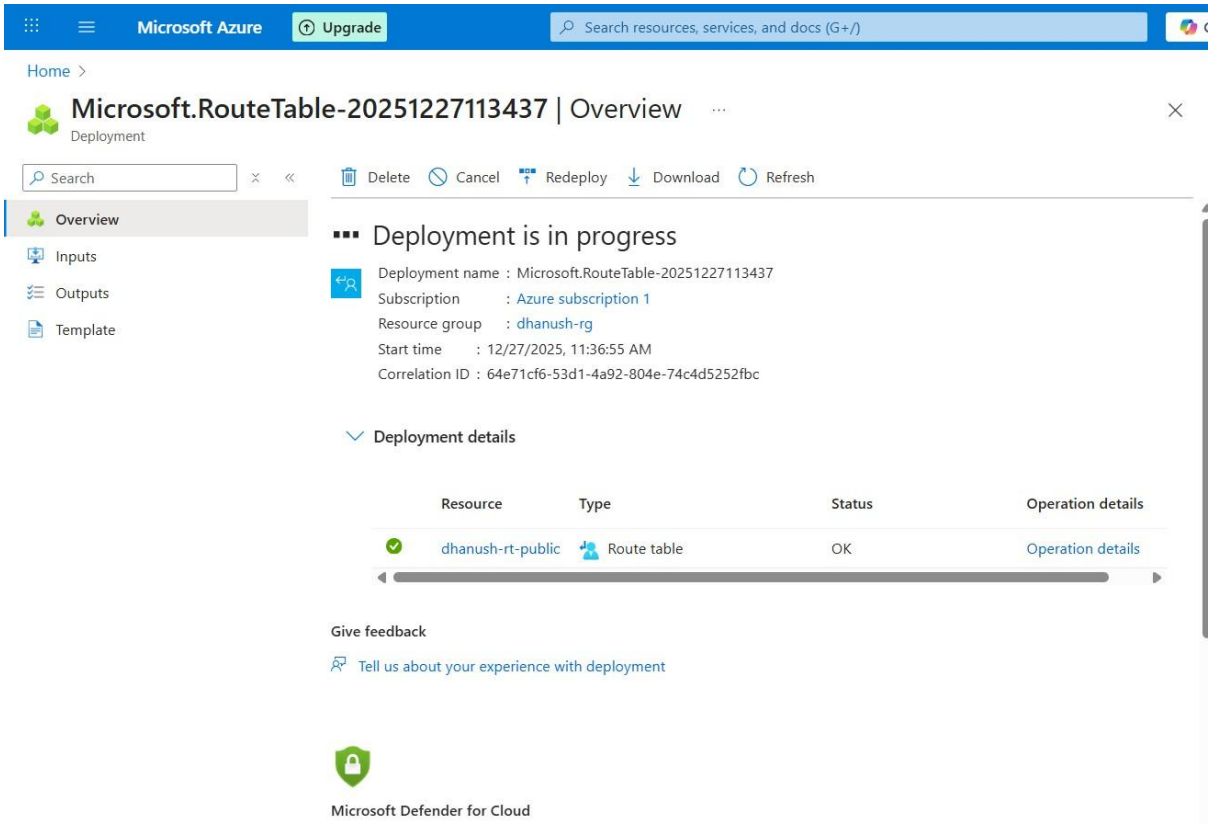
1. Go to Route tables → Create
   - Name: <studentname>-rt-public
   - Resource group: <studentname>-rg
2. After creation → Open Route Table → Routes → Add
   - Name: internet-route
   - Address prefix: 0.0.0.0/0

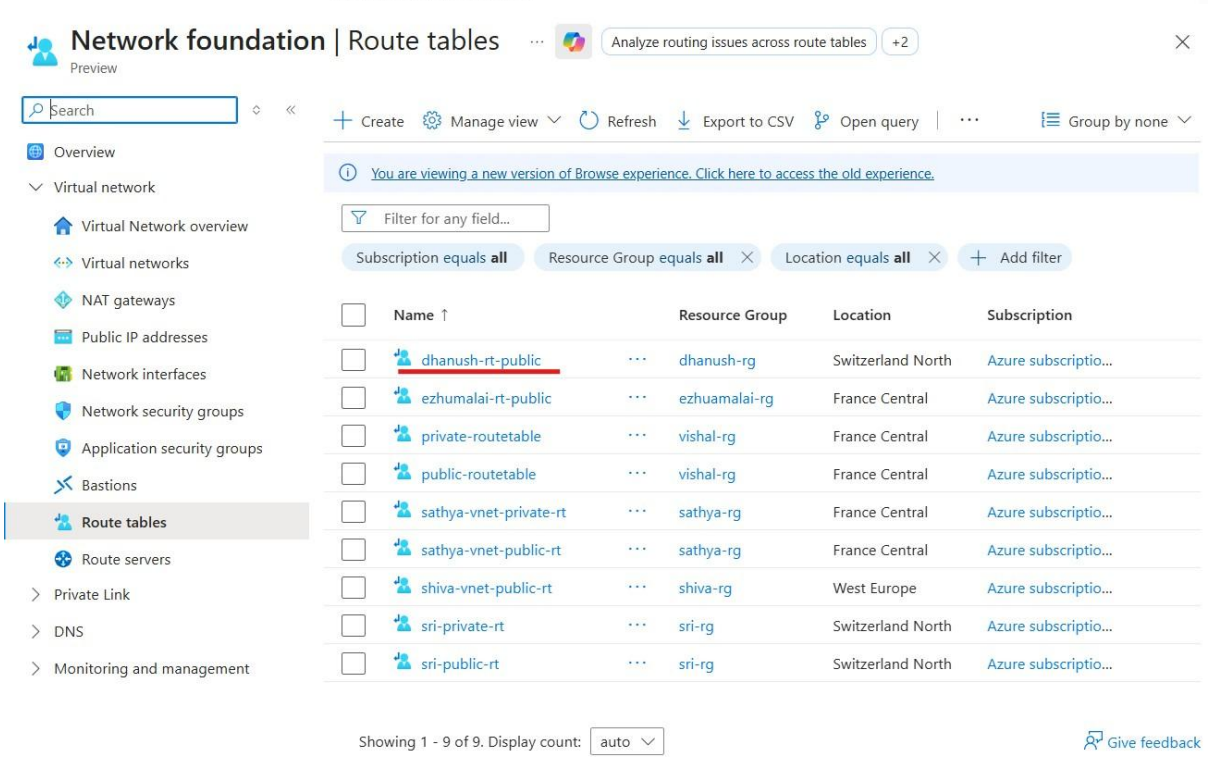
- Next hop: Internet

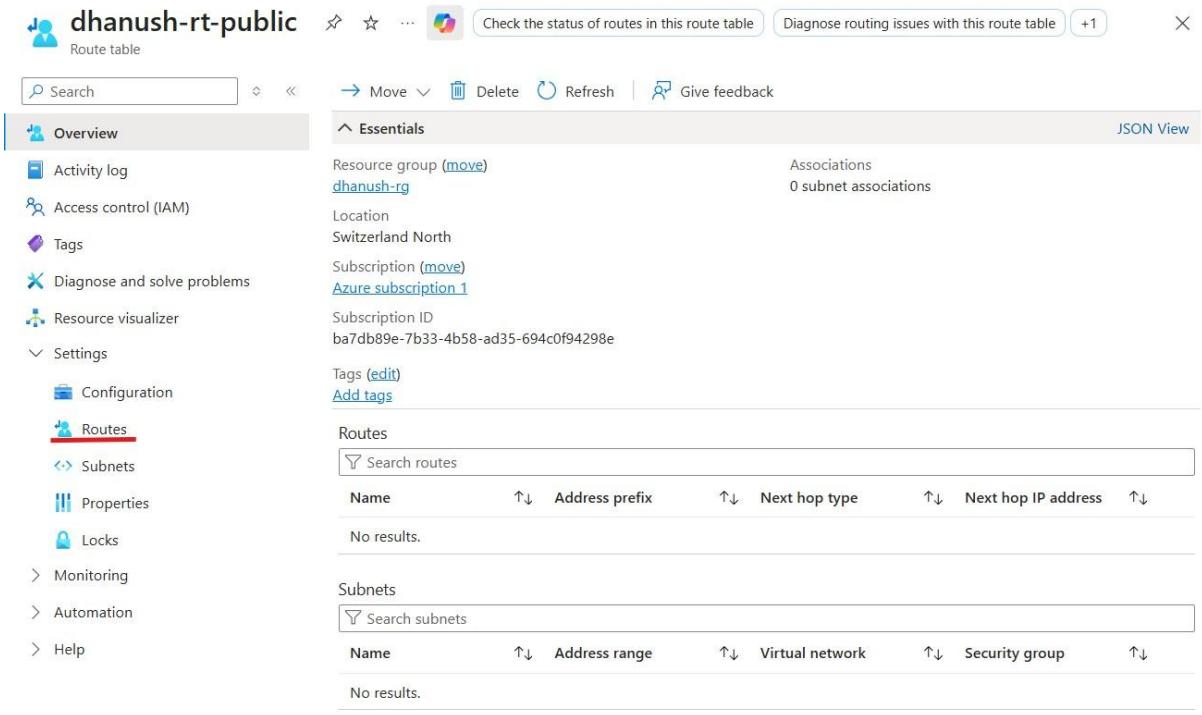
3. Associate subnet: <studentname>-public-subnet

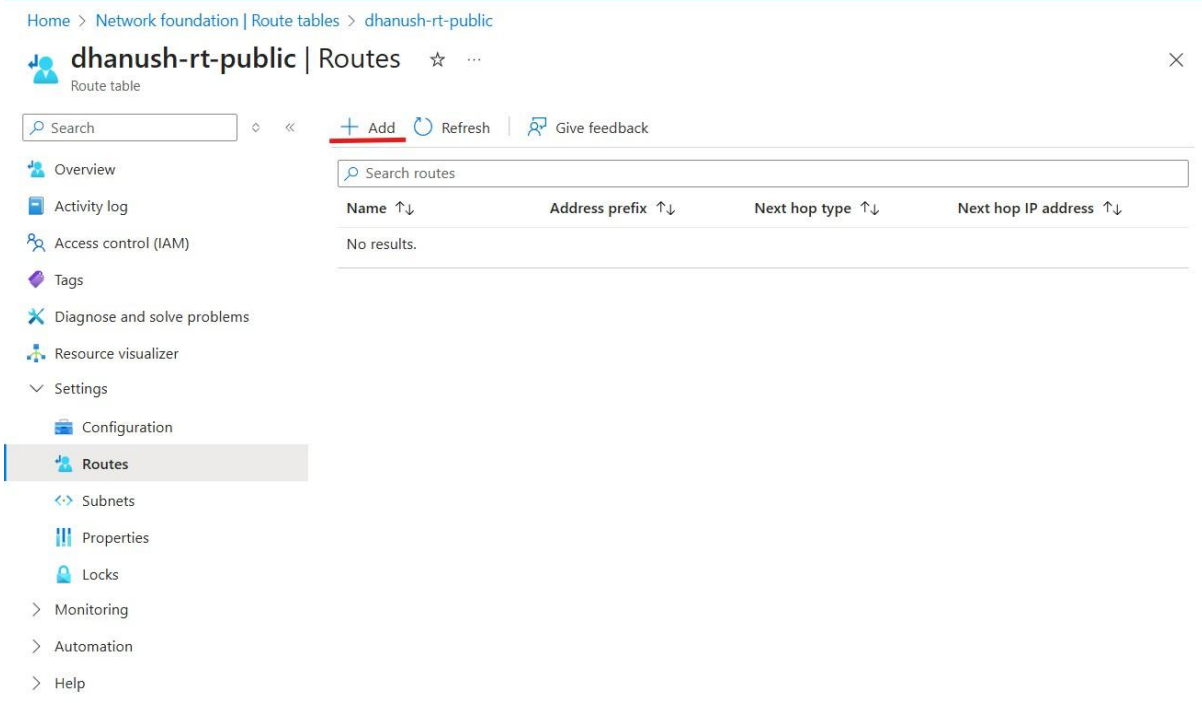


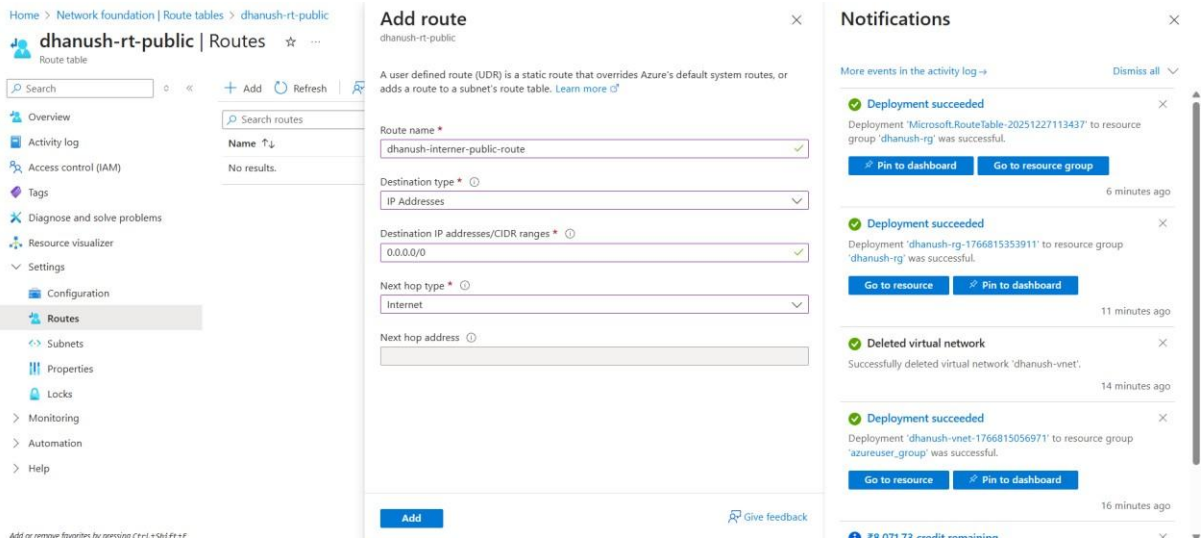

Microsoft Azure
Upgrade
Search resources, services, and docs (G+/)
Home > Network foundation | Route tables >
Create Route table
Basics
Tags
Review + create
View automation template
Basics
Subscription
Azure subscription 1
Resource group
dhanush-rg
Region
Switzerland North
Name
dhanush-rt-public
Propagate gateway routes
Yes
Previous
Next
Create
Give feedback

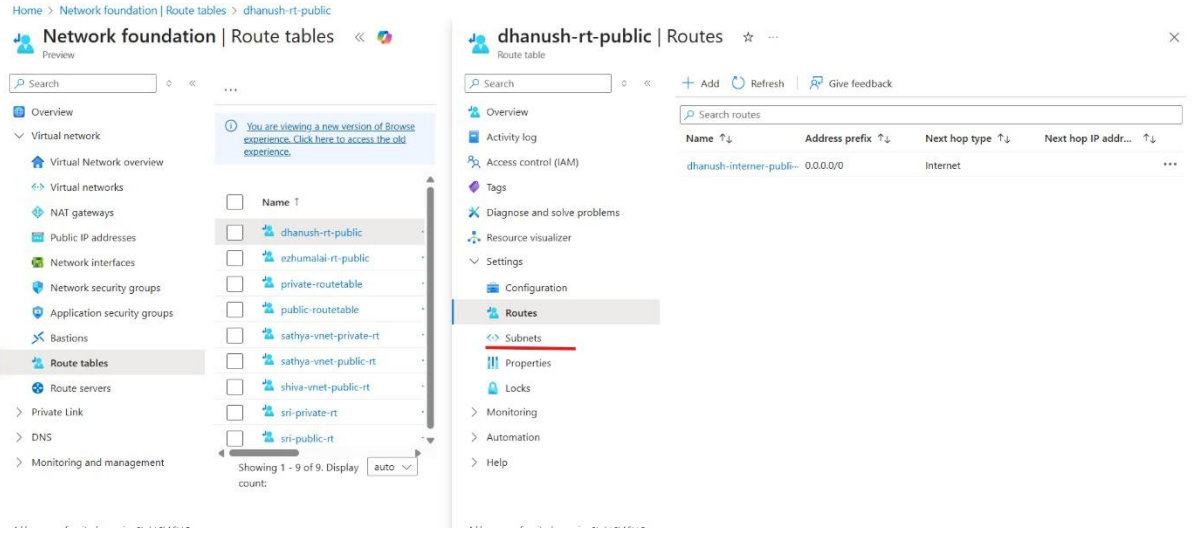

Microsoft Azure
Upgrade
Search resources, services, and docs (G+/)
Home
Microsoft.RouteTable-20251227113437 | Overview
Deployment
Search
Overview
Inputs
Outputs
Template
Delete
Cancel
Redeploy
Download
Refresh
Deployment is in progress
Deployment name : Microsoft.RouteTable-20251227113437
Subscription : Azure subscription 1
Resource group : dhanush-rg
Start time : 12/27/2025, 11:36:55 AM
Correlation ID : 64e71cf6-53d1-4a92-804e-74c4d5252fbc
Deployment details
Resource
Type
Status
Operation details
dhanush-rt-public
Route table
OK
Operation details
Give feedback
Tell us about your experience with deployment
Microsoft Defender for Cloud

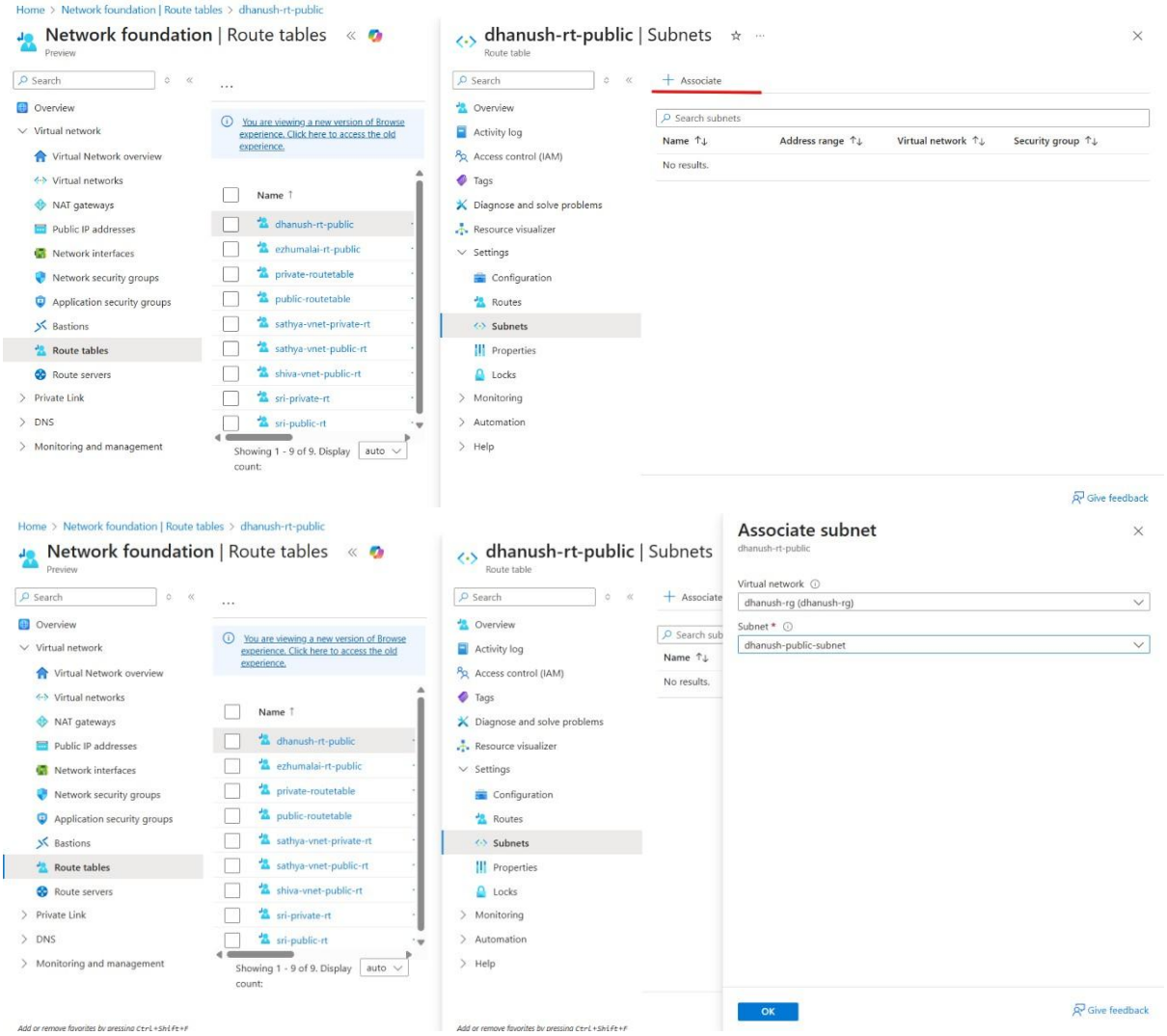

Network foundation | Route tables
Preview
Analyze routing issues across route tables
+2
Search
Overview
Virtual network
Virtual Network overview
Virtual networks
NAT gateways
Public IP addresses
Network interfaces
Network security groups
Application security groups
Bastions
Route tables
Route servers
Private Link
DNS
Monitoring and management
Create
Manage view
Refresh
Export to CSV
Open query
Group by none
You are viewing a new version of Browse experience. Click here to access the old experience.
Filter for any field...
Subscription equals all
Resource Group equals all
Location equals all
Add filter
Name
Resource Group
Location
Subscription
dhanush-rt-public
dhanush-rg
Switzerland North
Azure subscriptio...
ezhumalai-rt-public
ezhuamalai-rg
France Central
Azure subscriptio...
private-routetable
vishal-rg
France Central
Azure subscriptio...
public-routetable
vishal-rg
France Central
Azure subscriptio...
sathya-vnet-private-rt
sathya-rg
France Central
Azure subscriptio...
sathya-vnet-public-rt
sathya-rg
France Central
Azure subscriptio...
shiva-vnet-public-rt
shiva-rg
West Europe
Azure subscriptio...
sri-private-rt
sri-rg
Switzerland North
Azure subscriptio...
sri-public-rt
sri-rg
Switzerland North
Azure subscriptio...
Showing 1 - 9 of 9. Display count: auto
Give feedback


dhanush-rt-public
Route table
Check the status of routes in this route table
Diagnose routing issues with this route table
+1
Search
Move
Delete
Refresh
Give feedback
Overview
Activity log
Access control (IAM)
Tags
Diagnose and solve problems
Resource visualizer
Settings
Configuration
Routes
Subnets
Properties
Locks
Monitoring
Automation
Help
Essentials
JSON View
Resource group (move)
dhanush-rg
Location
Switzerland North
Subscription (move)
Azure subscription 1
Subscription ID
ba7db89e-7b33-4b58-ad35-694c0f94298e
Tags (edit)
Add tags
Associations
0 subnet associations
Routes
Search routes
Name
Address prefix
Next hop type
Next hop IP address
No results.
Subnets
Search subnets
Name
Address range
Virtual network
Security group
No results.


Home > Network foundation | Route tables > dhanush-rt-public
dhanush-rt-public | Routes
Route table
Search
Add
Refresh
Give feedback
Overview
Activity log
Access control (IAM)
Tags
Diagnose and solve problems
Resource visualizer
Settings
Configuration
Routes
Subnets
Properties
Locks
Monitoring
Automation
Help
Search routes
Name
Address prefix
Next hop type
Next hop IP address
No results.


Home > Network foundation | Route tables > dhanush-rt-public
dhanush-rt-public | Routes
Route table
Search
Overview
Activity log
Access control (IAM)
Tags
Diagnose and solve problems
Resource visualizer
Settings
Configuration
Routes
Subnets
Properties
Locks
Monitoring
Automation
Help
Add
Refresh
Search routes
Name
No results.
Add route
dhanush-rt-public
A user defined route (UDR) is a static route that overrides Azure's default system routes, or adds a route to a subnet's route table. Learn more
Route name *
dhanush-interner-public-route
Destination type *
IP Addresses
Destination IP addresses/CIDR ranges *
0.0.0.0/0
Next hop type *
Internet
Next hop address
Add
Give feedback
Notifications
More events in the activity log
Dismiss all
Deployment succeeded
Deployment 'Microsoft.RouteTable-20251227113437' to resource group 'dhanush-rg' was successful.
Pin to dashboard
Go to resource group
6 minutes ago
Deployment succeeded
Deployment 'dhanush-rg-1766815353911' to resource group 'dhanush-rg' was successful.
Go to resource
Pin to dashboard
11 minutes ago
Deleted virtual network
Successfully deleted virtual network 'dhanush-vnet'.
14 minutes ago
Deployment succeeded
Deployment 'dhanush-vnet-1766815056971' to resource group 'azureuser_group' was successful.
Go to resource
Pin to dashboard
16 minutes ago

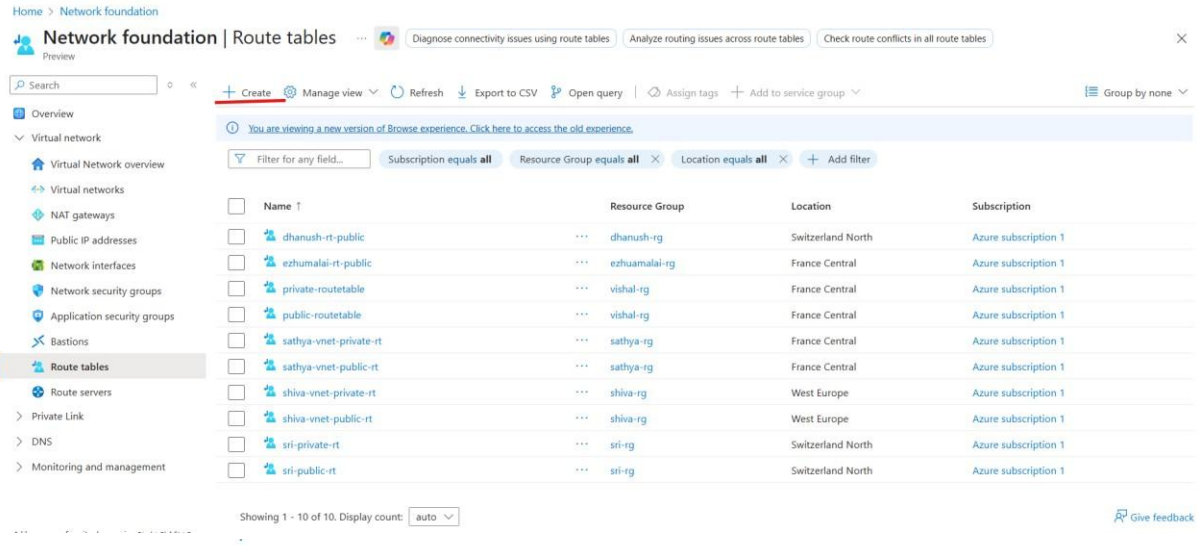

Home > Network foundation | Route tables > dhanush-rt-public
Network foundation | Route tables
Preview
Search
Overview
Virtual network
Virtual Network overview
Virtual networks
NAT gateways
Public IP addresses
Network interfaces
Network security groups
Application security groups
Bastions
Route tables
Route servers
Private Link
DNS
Monitoring and management
You are viewing a new version of Browse experience. Click here to access the old experience.
Name
dhanush-rt-public
ezhumalai-rt-public
private-routetable
public-routetable
sathya-vnet-private-rt
sathya-vnet-public-rt
shiva-vnet-public-rt
sri-private-rt
sri-public-rt
Showing 1 - 9 of 9. Display count:
auto
dhanush-rt-public | Routes
Route table
Search
Overview
Activity log
Access control (IAM)
Tags
Diagnose and solve problems
Resource visualizer
Settings
Configuration
Routes
Subnets
Properties
Locks
Monitoring
Automation
Help
Add
Refresh
Give feedback
Search routes
Name
Address prefix
Next hop type
Next hop IP addr...
dhanush-interner-publi...
0.0.0.0/0
Internet

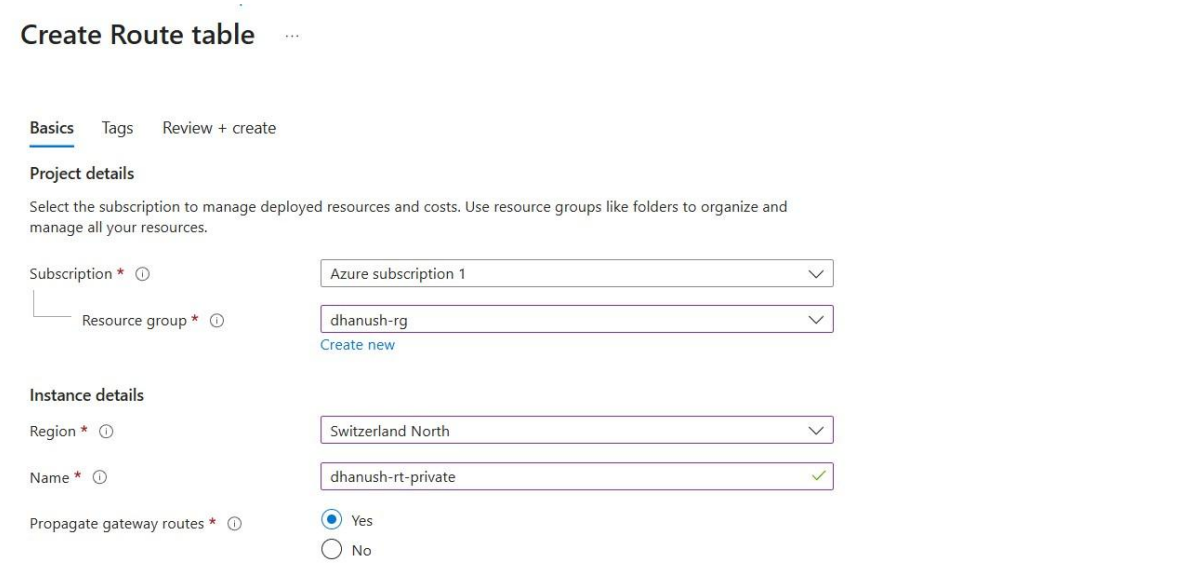


Private Route Table

1. Create another route table:
   - Name: <studentname>-rt-private
   - Resource group: <studentname>-rg
2. No new routes needed (default local routes apply)

3. Associate subnet: <studentname>-private-subnet





Create Route table ...

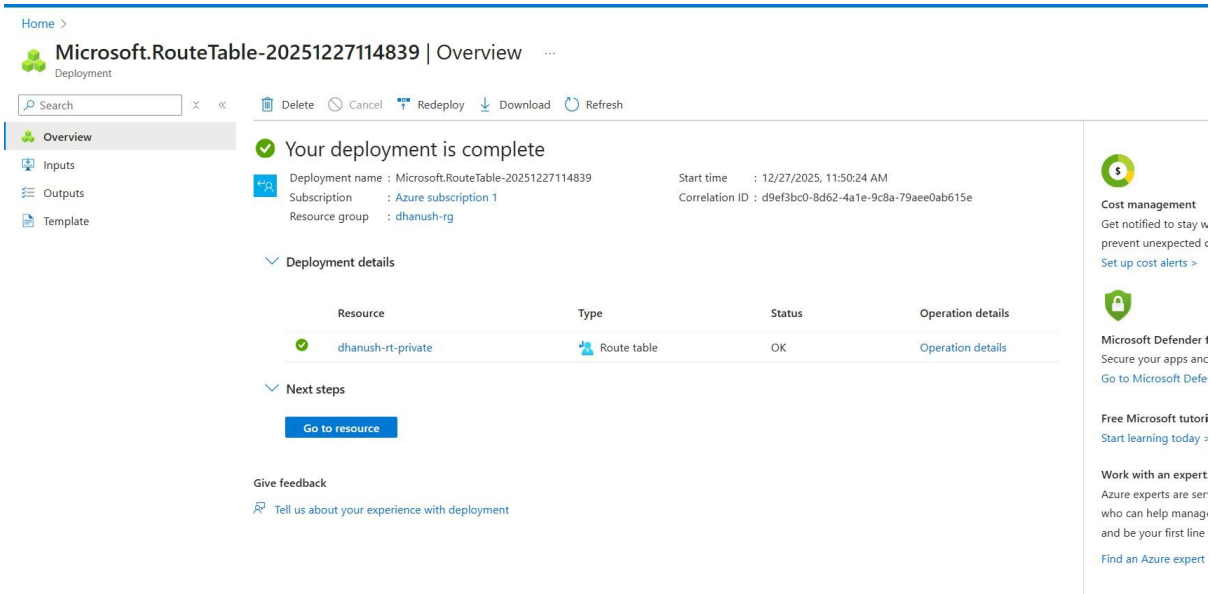
Basics Tags **Review + create**

View automation template

Basics

Subscription	Azure subscription 1
Resource group	dhanush-rg
Region	Switzerland North
Name	dhanush-rt-private
Propagate gateway routes	Yes

Previous Next **Create**



STEP 4: Create Jumpbox VM (Public Access)

1. Go to Virtual machines → Create
    - Name: <studentname>-jumpbox
    - Image: AlmaLinux 9
    - Size: Standard_B1s
    - Authentication: SSH key or password
    - Networking:
        - VNet: <studentname>-vnet
        - Subnet: <studentname>-public-subnet
        - Public IP: Enabled
        - NIC NSG: Allow SSH (22) inbound
2. Review + Create → Create

Subscription * ⓘ: Azure subscription 1

Resource group * ⓘ: dhanush-rg

Create new

Instance details

Azure detected there are other region, size and zone recommendations that may lead to a higher chance of successful deployments. To deploy seamlessly, please adopt any of the suggested recommendations. View recommendation.

Virtual machine name * ⓘ: dhanush-vm

Region * ⓘ: (Europe) Switzerland North

Deploy to an Azure Extended Zone

Availability options ⓘ: Availability zone

Deploy to an Azure Extended Zone

Availability options ⓘ: Availability zone

Zone options ⓘ:
- (●) Self-selected zone — Choose up to 3 availability zones, one VM per zone
- ( ) Azure-selected zone (Preview) — Let Azure assign the best zone for your needs

Availability zone * ⓘ: Zone 1

You can now select multiple zones. Selecting multiple zones will create one VM per zone. Learn more

Security type ⓘ: Trusted launch virtual machines

Configure security features

Image * ⓘ: AlmaLinux OS 9 - x64 Gen2 (free services eligible)

See all images | Configure VM generation

VM architecture ⓘ:
- ( ) Arm64
- (●) x64

Administrator account

Authentication type ⓘ:
- ( ) SSH public key
- (●) Password

Username * ⓘ: dhanushuser

Password *: ••••••••••••••

Confirm password *: ••••••••••••••

Inbound port rules

Select which virtual machine network ports are accessible from the public internet. You can specify more limited or granular network access on the Networking tab.

Public inbound ports * ⓘ:
- ( ) None
- (●) Allow selected ports

Select inbound ports *: SSH (22)

Basics Disks **Networking** Management Monitoring Advanced Tags Review + create

Define network connectivity for your virtual machine by configuring network interface card (NIC) settings. You can control ports, inbound and outbound connectivity with security group rules, or place behind an existing load balancing solution.
Learn more

Network interface

When creating a virtual machine, a network interface will be created for you.

Virtual network * : dhanush-rg
Create new

Subnet * : dhanush-public-subnet (10.0.1.0/24)
Manage subnet configuration

Public IP : (new) dnaush-vm-ip
Create new
Public IP addresses have a nominal charge. Estimate price

NIC network security group : None / Basic / Advanced

Create a virtual machine

Help me create a low cost VM | Help me choose the right VM size for my workload | Help me create a VM optimized for high availability

Validation passed

Help me create a low cost VM | Help me create a VM optimized for high availability | Help me choose the right VM size for my workload

Basics Disks Networking Management Monitoring Advanced Tags **Review + create**

Price

1 X Standard D2s v3
by Microsoft
Terms of use | Privacy policy

Subscription credits apply
0.1320 USD/hr
Pricing for other VM sizes

TERMS

By clicking "Create", I (a) agree to the legal terms and privacy statement(s) associated with the Marketplace offering(s) listed above; (b) authorize Microsoft to bill my current payment method for the fees associated with the offering(s), with the same billing frequency as my Azure subscription; and (c) agree that Microsoft may share my contact, usage and transactional information with the provider(s) of the offering(s) for support, billing and other transactional activities. Microsoft does not provide rights for third-party offerings. See the Azure Marketplace Terms for additional details.

Name: dhanush v
Preferred e-mail address: dhanushnovember05@gmail.com

< Previous | Next > | Create

Download a template for

CreateVm-almalinux.almalinux-x86_64-9-gen2-20251227123259 | Overview

Deployment

Search

Overview
Inputs
Outputs
Template

Delete | Cancel | Redeploy | Download | Refresh

Deployment is in progress

Deployment name: CreateVm-almalinux.almalinux-x86_64-9-gen2-...
Subscription: Azure subscription 1
Resource group: dhanush-rg

Start time: 12/27/2025, 12:36:05 PM
Correlation ID: 793db59c-43fd-437f-85d6-cf9ec636b46a

Deployment details

Resource	Type	Status	Operation details
dnaush-vm433_z1	Microsoft.Network/networkInterfa...	OK	Operation details
dnaush-vm-nsg	Microsoft.Network/networkSecuri...	OK	Operation details
dnaush-vm-ip	Microsoft.Network/publicIpAddre...	OK	Operation details

Give feedback
Tell us about your experience with deployment

Microsoft
Secure yo
Go to Mic

Free Micr
Start learn

Work with
Azure exp
who can h
and be yo
Find an Az

STEP 5: Create Private VM

1. Go to Virtual machines → Create
    - Name: <studentname>-private-vm
    - Image: AlmaLinux 9
    - Size: Standard_B1s
    - Authentication: SSH key or password
    - Networking:
        - VNet: <studentname>-vnet
        - Subnet: <studentname>-private-subnet
        - Public IP: None
        - NIC NSG: Allow SSH (22) only from VNet
2. Review + Create → Create

Subscription * ⓘ — Azure subscription 1

Resource group * ⓘ — dhanush-rg

Create new

Instance details

Virtual machine name * ⓘ — dhanush-vm2

Region * ⓘ — (Europe) Switzerland North

Deploy to an Azure Extended Zone

Availability options ⓘ — Availability zone

Zone options ⓘ

- (•) Self-selected zone
  Choose up to 3 availability zones, one VM per zone
- ( ) Azure-selected zone (Preview)
  Let Azure assign the best zone for your needs

Availability zone * ⓘ — Zone 1

You can now select multiple zones. Selecting multiple zones will create one VM per zone. Learn more

< Previous | Next : Disks > | Review + create

Security type ⓘ — Trusted launch virtual machines

Configure security features

Image * ⓘ — AlmaLinux OS 9 - x64 Gen2 (free services eligible)

See all images | Configure VM generation

VM architecture ⓘ

- ( ) Arm64
- (•) x64

Arm64 is not supported with the selected image.

Run with Azure Spot discount ⓘ — [ ]

You are in the free trial period. Costs associated with this VM can be covered by any remaining credits on your subscription. Learn more

Size * ⓘ — Standard_D2s_v3 - 2 vcpus, 8 GiB memory ($96.36/month)

See all sizes

Administrator account

Authentication type ⓘ

- ( ) SSH public key
- (•) Password

Username * ⓘ — dhansuhuser

Password * — ••••••••••••••

Confirm password * — ••••••••••••••

Inbound port rules

Select which virtual machine network ports are accessible from the public internet. You can specify more limited or granular network access on the Networking tab.

Public inbound ports * ⓘ

- ( ) None
- (•) Allow selected ports

Select inbound ports * — HTTP (80), SSH (22)

This will allow all IP addresses to access your virtual machine. This is only

Help me create a low cost VM | Help me create a VM optimized for high availability | Help me choose the right VM size for my workload

Network interface

When creating a virtual machine, a network interface will be created for you.

Virtual network * : dhanush-rg
Create new

Subnet * : dhanush-private-subnet (10.0.2.0/24)
Manage subnet configuration

Public IP : None
Create new

NIC network security group : None / Basic / Advanced

Public inbound ports * : None / Allow selected ports

Select inbound ports * : HTTP (80), SSH (22)

Create a virtual machine

Validation passed

Help me create a low cost VM | Help me create a VM optimized for high availability | Help me choose the right VM size for my workload

Basics | Disks | Networking | Management | Monitoring | Advanced | Tags | **Review + create**

Price

1 X Standard D2s v3
by Microsoft
Terms of use | Privacy policy

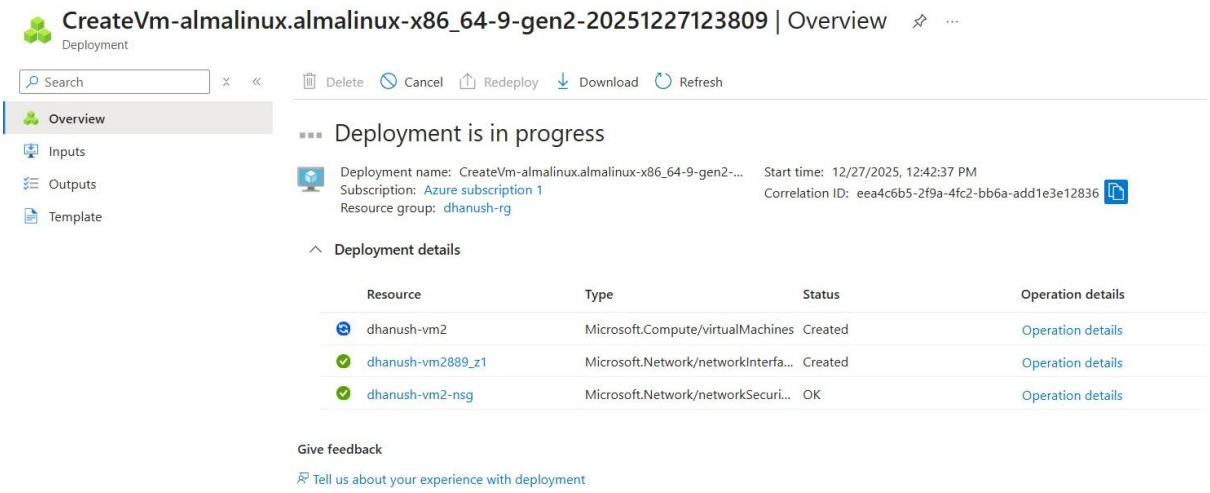
Subscription credits apply
0.1320 USD/hr
Pricing for other VM sizes

TERMS

By clicking "Create", I (a) agree to the legal terms and privacy statement(s) associated with the Marketplace offering(s) listed above; (b) authorize Microsoft to bill my current payment method for the fees associated with the offering(s), with the same billing frequency as my Azure subscription; and (c) agree that Microsoft may share my contact, usage and transactional information with the provider(s) of the offering(s) for support, billing and other transactional activities. Microsoft does not provide rights for third-party offerings. See the Azure Marketplace Terms for additional details.

Name: dhanush v

Preferred e-mail address: dhanushnovember05@gmail.com



STEP 6: Test Access

1. From your local PC → SSH into jumpbox:

ssh azureuser@<jumpbox-public-ip>

2. Inside jumpbox → SSH into private VM using private IP:

ssh azureuser@10.0.2.4

(replace with actual private IP)

```
C:\Users\DHANUSH>ssh dhanushuser@20.199.128.161
The authenticity of host '20.199.128.161 (20.199.128.161)' can't be established.
ED25519 key fingerprint is SHA256:tkDHGinrgMf2fIfA1ala6LxC1elzVYhxANDYvf5pYlo.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '20.199.128.161' (ED25519) to the list of known hosts.
dhanushuser@20.199.128.161's password:
[dhanushuser@dnaush-vm ~]$ sudo -i
[root@dnaush-vm ~]# hostname
dnaush-vm
[root@dnaush-vm ~]# curl ifconfig.me
20.199.128.161[root@dnaush-vm ~]# hostname -I
10.0.1.4
[root@dnaush-vm ~]# ssh dhanushuser@10.0.1.4
The authenticity of host '10.0.1.4 (10.0.1.4)' can't be established.
ED25519 key fingerprint is SHA256:tkDHGinrgMf2fIfA1ala6LxC1elzVYhxANDYvf5pYlo.
This key is not known by any other names
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '10.0.1.4' (ED25519) to the list of known hosts.
dhanushuser@10.0.1.4's password:
Last login: Sat Dec 27 07:24:39 2025 from 122.172.87.126
[dhanushuser@dnaush-vm ~]$
```

Checking public ip : curl ifconfig.me

Checking private ip : hostname -I

End Result

- <studentname>-jumpbox → Public access (via public subnet + route)
- <studentname>-private-vm → No internet access, only reachable from jumpbox
- Route tables correctly applied:
  - Public RT → internet route 0.0.0.0/0
  - Private RT → local only